Weekly Meeting 3

LGAD reading and breakdown simulation update

Prof. Prabhakar Palni group meeting

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Focus: two papers, one TCAD theory point, and breakdown runs

Plan for today

1. Reading from two recent LGAD papers.
2. One TCAD theory point: incomplete ionization in 4H-SiC.
3. What went wrong in my first 4H-SiC PIN breakdown attempt.
4. New silicon PIN and LGAD breakdown simulations.
5. What I need to check next.

Scope This is still a learning and debugging update. I am trying to move from just running decks to understanding which results are physically meaningful.

Review paper: LGAD design directions

- The review paper gave me a broader map of LGAD R&D.
- I saw several design directions:
  - inverse LGADs (iLGAD)
  - trench-isolated LGADs
  - AC- and DC-coupled resistive LGADs
  - compensated LGADs for high fluence
- The common goal is still the same: keep gain and timing while improving segmentation and radiation tolerance.

Reading point I did not read the review paper in full depth yet, but it helped me place AC-LGAD, TI-LGAD, iLGAD, and compensated LGADs in one picture.

Review paper: thin LGAD timing

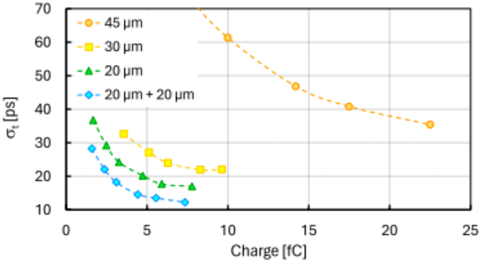


FIGURE 1
Timing resolution as a function of the collected charge from EXFLU1 LGAD sensors with different thicknesses: 45 µm (orange circles), 30 µm (yellow squares), 20 µm (green triangles), and two 20 µm sensors (blue diamonds). The error bars are not displayed.

- The paper emphasizes very thin LGADs for 4D tracking.
- Thinner active layers reduce the non-uniform ionization contribution.
- The best results are in the 20–30 μm range, and combining two 20 μm sensors improves timing further.

ONSEMI paper: 4H-SiC LGAD structure

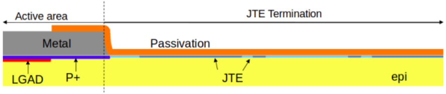


Figure 2. Illustration of the full 4H-SiC LGAD detector design, including the epitaxial layer, gain layer, and junction termination extension for a reasonably high breakdown voltage.

Top	metal/readout and passivation
Near surface	LGAD gain layer and p^{+} implant
Edge	JTE region to reduce edge breakdown
Bulk	4H-SiC epitaxial layer

- The device uses a 4H-SiC epitaxial layer.
- The gain layer is placed close to the top side.
- Junction termination extension (JTE) is included to support high breakdown voltage.
- This is closer to the real structure we eventually need to model.

Simulation link An LGAD deck should include the gain layer and edge termination/JTE.

ONSEMI paper: measurements that can become targets

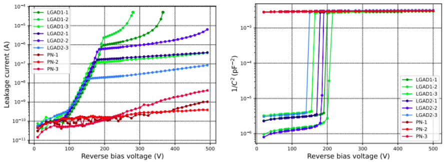


Figure 4. Measured IV (left) and CV (right) curves of the manufactured sensors from the 3 wafers with 30 µm thick epitaxial layer and different parameters of epitaxial implantation and gain layer. The depletion of the gain layer occurs between 155 V and 220 V, aligning well with TCAD simulation predictions.

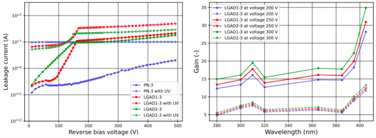


Figure 5. Response to UV light: Leakage current of the sensors measured for the wavelength of 277 nm (left) and gain as a function of wavelength (right).

- Figure 4 gives measured IV and CV curves for PN and LGAD sensors.
- The gain-layer depletion is reported around 155–220 V.
- Figure 5 connects leakage-current response and gain under UV illumination.

Use in TCAD The first useful checks are IV shape, CV/gain-layer depletion voltage, and then gain response.

TCAD theory point: incomplete ionization

$$N_A^- = \frac{N_A}{1 + g_A \exp\left(\frac{E_A - E_F}{k_B T}\right)}$$

$$N_D^+ = \frac{N_D}{1 + g_D \exp\left(\frac{E_F - E_D}{k_B T}\right)}$$

- For Al in 4H-SiC: $g_A = 4$, $E_A \approx E_V + 0.20$ eV.
- For nitrogen donors: $g_D = 2$.
- At 300 K, only part of the Al acceptor concentration becomes active charge.

Practical TCAD consequence

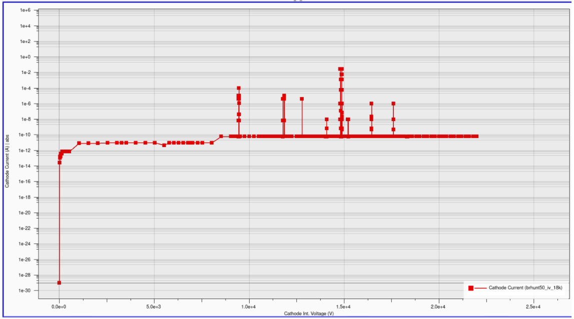
The deck should not treat raw Al doping as fully active charge in 4H-SiC.

Model direction

Add incomplete ionization and set Al ionization energy/degeneracy before serious SiC gain-layer runs.

This is one reason silicon was useful as a simpler debug material first.

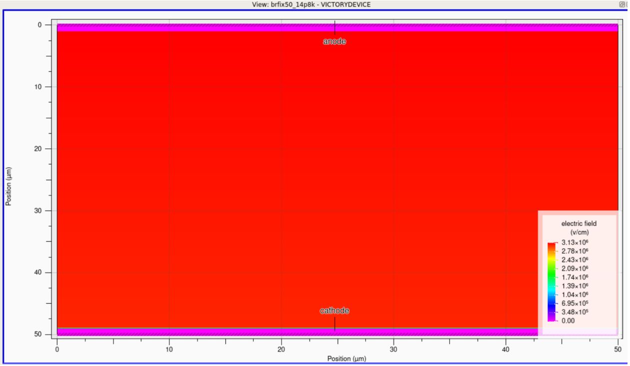
4H-SiC PIN breakdown attempt: not reliable yet



Problem The spikes look like solver instability, not physical breakdown.

- Large voltage steps pushed the solve into the post-breakdown region.
- No early current compliance, so Newton could jump to artificial high-current states.
- The SiC run still needs calibrated impact/material parameters and incomplete ionization.

4H-SiC PIN diagnostic field at high bias



- This was near the last saved high-bias state around 14.8 kV.
- The field scale reaches the MV/cm range.
- The plot is useful as a diagnostic, but not enough to claim a clean breakdown voltage.

Decision I shifted to silicon first to debug the breakdown workflow with better-known models.

What I changed for the silicon breakdown decks

- Switched to silicon to debug the breakdown workflow first.
- Used a silicon-specific avalanche model:

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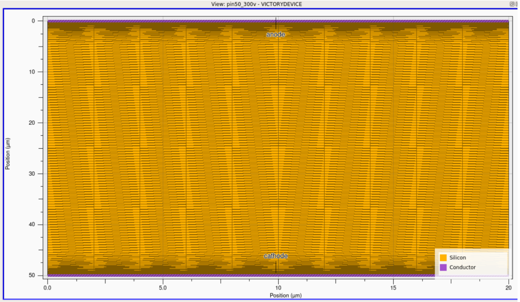
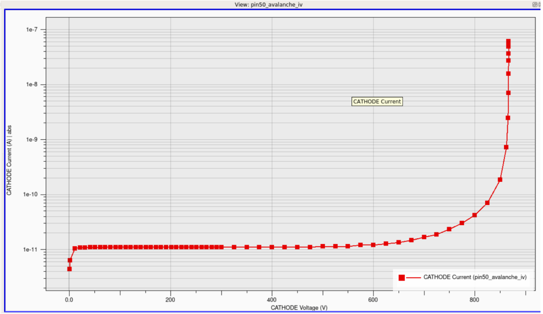
- Added a controlled SRH seed current using carrier lifetimes.
- Used smaller voltage steps near expected breakdown.
- Added current control so the solver stops before runaway.
- Saved impact generation and ionization-integral outputs for checking.

Result The silicon curves became smoother because the sweep was controlled before the post-breakdown region.

Silicon PIN: structure and avalanche IV



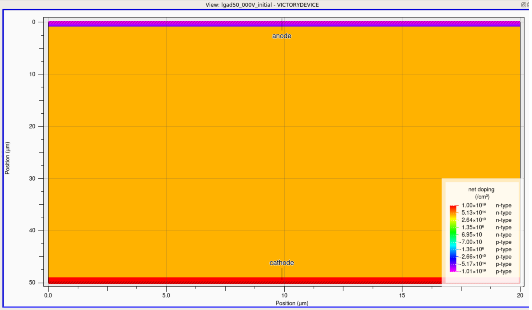
50 μm silicon PIN structure.



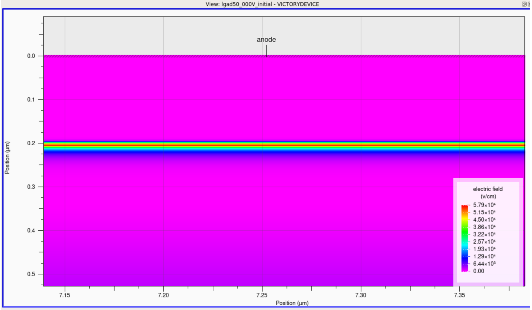
Cathode current rises sharply in the high-bias region.

What improved Compared with the failed SiC run, the silicon PIN sweep gives a cleaner avalanche-like current rise without the same random spike pattern.

Silicon LGAD: gain-layer setup

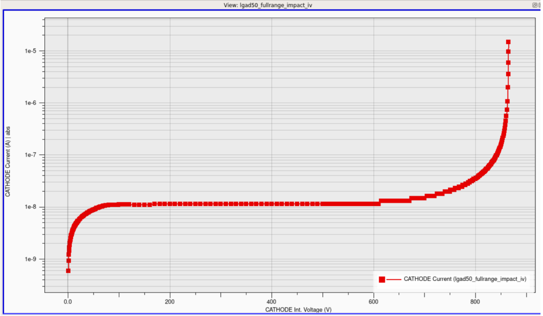


Net-doping view of the silicon LGAD-like structure.

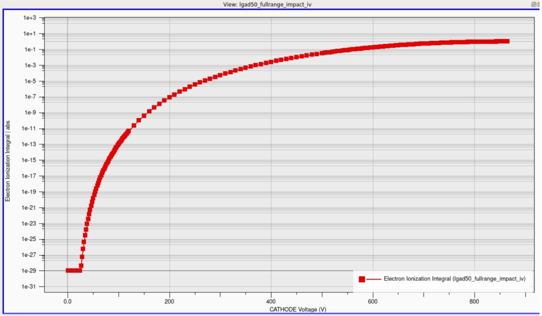


Zoomed electric-field view near the shallow layer.

Silicon LGAD: breakdown diagnostics



IV curve from the silicon LGAD-like deck.



Electron ionization integral rises toward and beyond unity.

Status This is still a model-debug result, but it is more interpretable than the earlier 4H-SiC breakdown attempt.

Next steps

- Clean the 4H-SiC PIN and LGAD decks.
- Save the useful fields and IV curves.
- Add a safer breakdown stopping condition.
- Try to find ONSEMI paper parameters and recreate the geometry with proper physics, including JTE.

Main direction Keep the next step small: one deck that is easier to explain and check.